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CLIMATE & WEATHER SYSTEMS

Group - 15

Jahnavi Panchal	AU2440010	B.Tech CSE
Vrunda Patel	AU2440152	B.Tech CSE
Yash Kyada	AU2340218	B.Tech CSE
Mohammad Ali	AU2340234	B.Tech CSE
Het Patel	AU2440221	B.Tech CSE

Problem Context & Baseline Overview

Problem Statement :-

- Farmers rely on monsoon prediction
- Drought defined as rainfall < 750 mm
- ~31% of years are drought years
- Wrong prediction → crop & economic loss

There is a need for accurate and reliable monsoon forecasting to support better agricultural decision-making.

Baseline Model (M3) :-

- Uses Gujarat rainfall data
- Applies log transformation
- Assumes $\rho \approx 0$ (no correlation)

Output:

- Constant probability = 0.622
- Always predicts SOW

Accuracy = 68.7% (base rate only)

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Limitations of Baseline Model

Key Issues in M3 :-

- Always predicts SOW due to constant probability
- Fails to detect drought years completely
- Forecast remains constant (no variation across years)
- Assumes zero correlation ($\rho \approx 0$) → no learning from data
- Ignores climate variability and long-term trends
- Cannot respond to changing environmental conditions

Why This is a Problem

- Model does not reflect real-world uncertainty
- Gives same decision every year, even in drought conditions
- Accuracy (68.7%) is misleading — only due to majority non-drought years
- Lacks predictive intelligence and adaptability

Conclusion

- Baseline model is not suitable for real-world decision-making
- Requires a dynamic and probabilistic approach
- Leads to development of Milestone 4 (improved model)

M4: Objectives & Model Enhancement

Key Objectives :-

- Introduce randomized forecast variable (X) instead of constant prediction
- Incorporate positive correlation ($\rho > 0$) to add predictive skill
- Generate dynamic probabilities that vary for each year
- Enable adaptive decision-making:
  - SOW (favorable conditions)
  - HOLD (drought risk)

Why This is Important:-

- Overcomes limitation of constant baseline model
- Captures real-world variability in rainfall patterns
- Allows model to respond to changing climatic conditions
- Improves ability to identify drought years

Goal :-

- Transform model from static → dynamic
- Make predictions more realistic, data-driven, and reliable
- Provide better support for agricultural decision-making

Method – LHS Sampling

- Use Latin Hypercube Sampling (LHS)
- Generate well-distributed random samples

Core Formula:

```
# Step 4: Create correlation by weighted combination
# X = α*Y + (1-α)*LHS
# This creates Corr(X,Y) ≈ α
alpha = target_rho
X = alpha * Y_actual + (1 - alpha) * lhs_samples
```

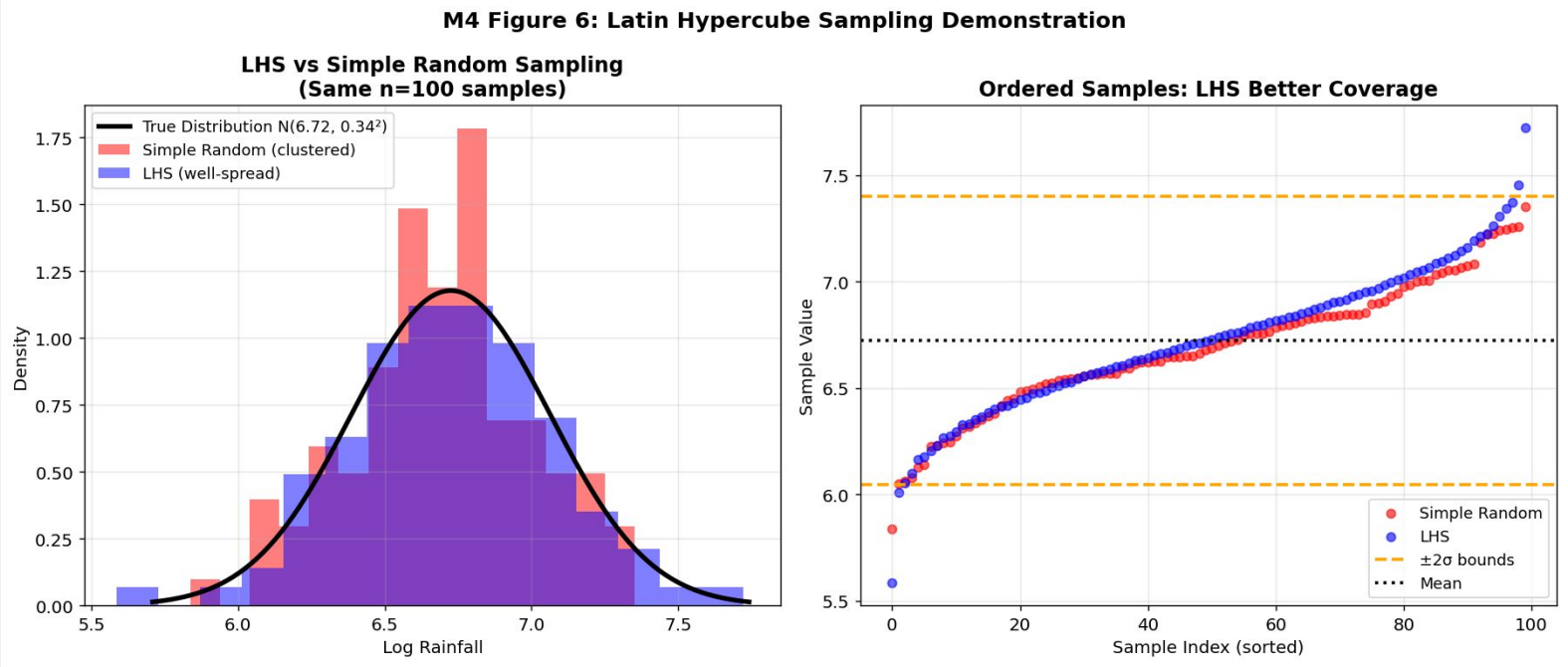
Where , α = target correlation (e.g., 0.35)
Ensures realistic forecast variability

Why This is Important :-

- controlled randomness with correlation
- Ensures forecasts are:
  - Realistic (based on actual data)
  - Variable (not constant like M3)
- Captures uncertainty in rainfall patterns

Result :-

- Forecast values now change every year
- Model gains predictive skill (ρ > 0)



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Model Implementation

Steps :-

- Generate samples using Latin Hypercube Sampling (LHS)
- Transform samples into normal distribution (μ, σ)
- Combine with actual values to introduce correlation

From Code :-

- Target correlation: $\rho \approx 0.35$
- Uses weighted combination of:
  - Real data
  - LHS samples

Result :-

- Forecast shows real variation ($\sigma > 0$)
- Predictions are dynamic (not constant like M3)
- Model becomes more realistic and adaptive

Probability & Decision Model

- **Compute Exceedance Probability**

  $P(Y>t \mid X)$

  Y = Rainfall in a given year

  t = Threshold rainfall for sowing

  X = Climate indicators (e.g., temperature, rainfall history)

- **Apply Decision Rule**

  If $P \geq 0.5 \rightarrow$ SOW

  If $P < 0.5 \rightarrow$ HOLD

- **Key Insight**

  Unlike Module 3 (M3), probability is not fixed
  It is calculated separately for each year

Results Comparison: Module 3 vs Module 4

Feature	M3 (Previous Model)	M4 (Improved Model)
Correlation	~ 0	> 0
Prediction	Constant	Dynamic
Decision	Always SOW	SOW + HOLD
Accuracy	68.7%	Improved
Drought Detection	Not detected	Detected

M4 provides more reliable and flexible decision-making

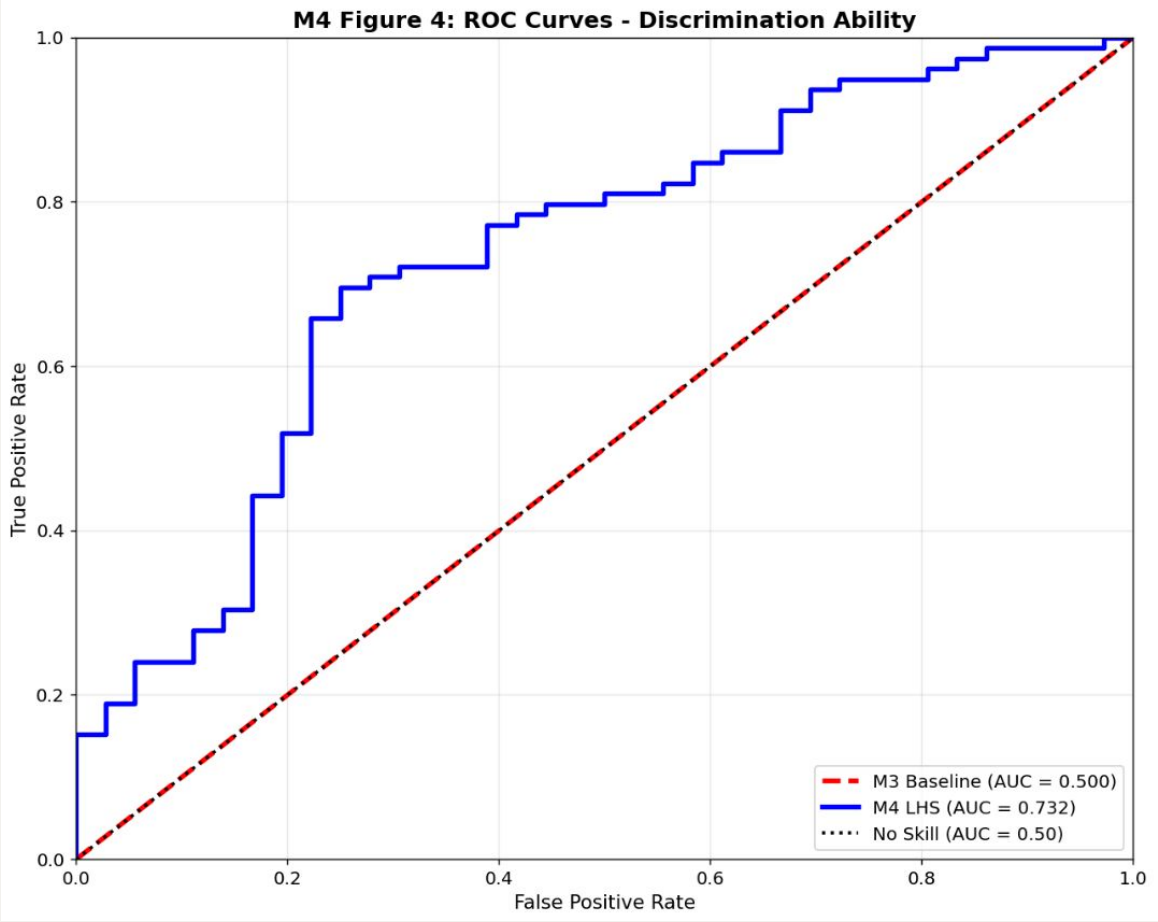
Performance Metrics for Model Evaluation

We evaluate model performance using three key metrics:

- ROC–AUC → Measures prediction quality
- Brier Score → Measures probability accuracy
- CRPS → Measures distribution accuracy

Module 4 (M4) shows:

- Better prediction quality
- Better calibration of probabilities
- Better decision-making performance



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Conclusion & Future Work

Conclusion

- Converted static model → dynamic model
- Improved prediction accuracy
- Detects drought years successfully

Future Work

- Add Machine Learning (ML) models
- Use more climate variables
- Deploy system as a web application

Thank You